Cambridge
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| CENTRE NUMBER | | CANDIDATE NUMBER | |

CAMBRIDGE
International Examinations

# 0DWKHPDWLFDO )RUPXODH

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### 4XDGUDWLF (TXDWLRQ

)RU WKH HTXDWLRQ D[ E[ F

$$[ \quad \frac{<E \pm \sqrt{E< \quad DF}}{D}$$

### %LQRPLDO 7KHRUHP

$$D \quad E^{Q} \quad D^{Q} \quad \binom{Q}{} D^{Q\pm} E \quad \binom{Q}{} D^{Q\pm} E \quad « \quad \binom{Q}{U} D^{Q\pm U} E^{U} \quad « \quad E^{Q}$$

ZKHUH Q LV D SRVLWLYH LQWHJHU DQG $\binom{Q}{U} \quad \frac{Q}{Q\pm U \; U}$

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### ,GHQWLWLHV

$$VL\$Q \quad FR\$V$$

$$VH\$F \quad WD\$Q$$

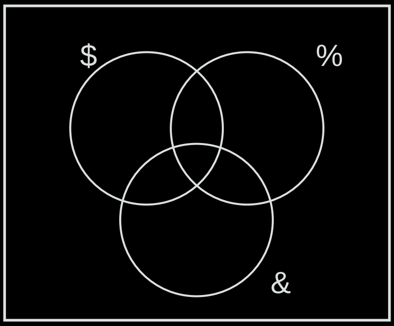
$$FRVH\$F \quad FR\$W$$

### )RUPXODH IRU %$%&

$$\frac{D}{VL\$Q} \quad \frac{E}{VL\%Q} \quad \frac{F}{VL\&Q}$$

$$D \quad E \quad F \pm EF \; FR\$V$$

$$¨ \quad – \; EF \; VL\$Q$$



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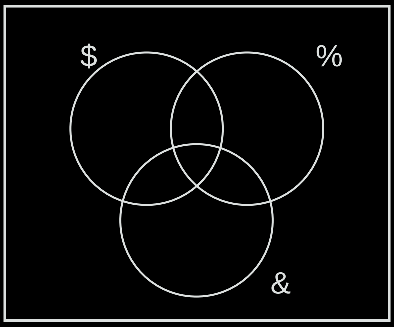

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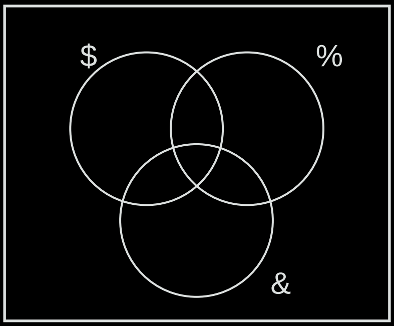

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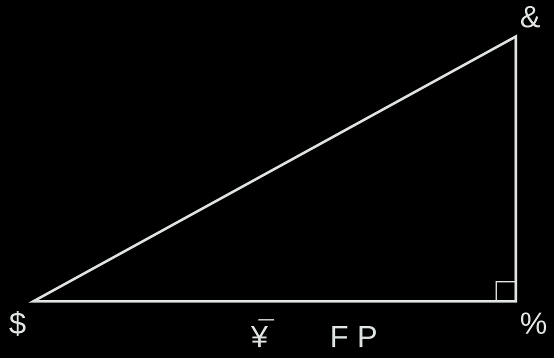
F * L Y H Q W K D W W < U L [ D D I L Q G W K H D O R X H I V R Z K L G K H W > @



7KHGLDJUVKRPZVWULD$Q%JOHWKGH √ FP$QJO%HVDULJKQWJ,OWHVJLYHQ

WKWDKWDHURIDWKWLULDLQJFOH

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LL+HQFHLQGKOHHQRJWKVH&LQWKRUS√ FPZKHSLHVQQWHJHU >@

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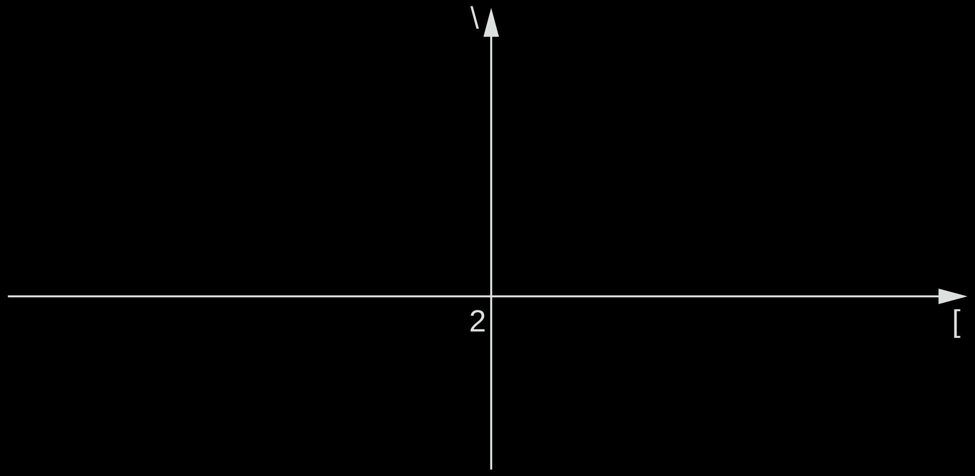

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LLL) LQG WKH YDOXHV RI WKH FRQVWDQW IRU ZKLFK WKH HTXDWLRQ | | KDV QR VROXWLRQV

[ D Q\GL W X FWK K D W *L Y H Q

W KD W FX UY DWX U D GL X V R I D SR L Q W R Q W KH FX UY H L Q W I L Q G WKH T X D G U DWLF H T X DWL R Q > @

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E 7R FRPSOHWH WKH TXL] WHDP LW EH FKRVHQ DPRQJ URXQG ZRPHQ )LQG WKH
QXPEHU RI GLIIHUHQW WHDPV WKDW FDQ EH FKRVHQ

L WKHUH DUH QR UHVWULFWLRQV > @

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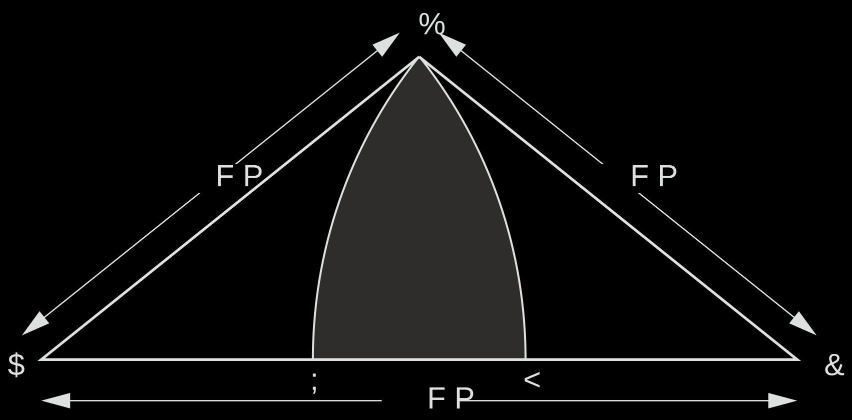

7KH GLDJUDP VKRZV DQ LVRVFHOHV WULDQJOH $%& ZKHUH $% %& FP DQG $& FP
%; LV DQ DUF RI D FLUFOH FHQWUH & DQG %< LV DQ DUF RI D FLUFOH FHQWUH $

L 6KRZ WKDW DQJOH $%& UDGLDQV FRUUHFW WR GHFLPDO SODFHV > @

LL )LQG WKH SHULPHWHU RI WKH VKDGHG UHJLRQ > @

LQWHUVHFWV DW SRLQWV $ DQG %. 7KH
SHUSHQGLFXODU ELVHFWRU RI $% LQWHUVHFWV WKH FLUFOH DW SRLQWV & DQG '. )LQG WKH OHQJWK RI WKH
OLQH &' LQ WKH IRUP √D ZKHUH D LV DQ LQWHJHU. > @

DQG — I L QWGKYHD ORXHHD RKW KLHQ W H[JDHQUGV> @

E 6ROYH ] ] > @